lab08

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Table of contents

Background 1
Preparing the Data 1
Applying PCA to the Data 5
Visualizing PCA Results 7
Variance Explained 10
Heirarchical Clustering 12
Combining Heirarchical Clustering with PCA 15
Predicting Diagnoses using PCA results 18

Background

Using the techniques learned throughout the class, we will be analyzing real breast cancer biopsy data through PCA.

Preparing the Data

load the data sets into R variables.

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names = 1)
head(wisc.df)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
84358402         M       20.29        14.34         135.10    1297.0
```

```
843786           M       12.45        15.70          82.57     477.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
84358402         0.10030          0.13280         0.1980             0.10430
843786           0.12780          0.17000         0.1578             0.08089
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
84358402        0.1809                0.05883    0.7572     0.7813        5.438
843786          0.2087                0.07613    0.3345     0.8902        2.217
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
84358402   94.44      0.011490        0.02461      0.05688           0.01885
843786     27.19      0.007510        0.03345      0.03672           0.01137
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
84358402     0.01756             0.005115        22.54         16.67
843786       0.02165             0.005082        15.47         23.75
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
84358402          152.20     1575.0           0.1374            0.2050
843786            103.40      741.6           0.1791            0.5249
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
84358402          0.4000               0.1625         0.2364
843786            0.5355               0.1741         0.3985
```

```
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
84358402                 0.07678
843786                   0.12440
```

removing diagnosis from the data frame and storing it in a separate variable diagnosis

```r
wisc.data <- wisc.df[,-1]
head(wisc.data)
```

```
         radius_mean texture_mean perimeter_mean area_mean smoothness_mean
842302         17.99        10.38         122.80    1001.0         0.11840
842517         20.57        17.77         132.90    1326.0         0.08474
84300903       19.69        21.25         130.00    1203.0         0.10960
84348301       11.42        20.38          77.58     386.1         0.14250
84358402       20.29        14.34         135.10    1297.0         0.10030
843786         12.45        15.70          82.57     477.1         0.12780
         compactness_mean concavity_mean concave.points_mean symmetry_mean
842302            0.27760         0.3001             0.14710        0.2419
842517            0.07864         0.0869             0.07017        0.1812
84300903          0.15990         0.1974             0.12790        0.2069
84348301          0.28390         0.2414             0.10520        0.2597
84358402          0.13280         0.1980             0.10430        0.1809
843786            0.17000         0.1578             0.08089        0.2087
         fractal_dimension_mean radius_se texture_se perimeter_se area_se
842302                  0.07871    1.0950     0.9053        8.589  153.40
842517                  0.05667    0.5435     0.7339        3.398   74.08
84300903                0.05999    0.7456     0.7869        4.585   94.03
84348301                0.09744    0.4956     1.1560        3.445   27.23
84358402                0.05883    0.7572     0.7813        5.438   94.44
843786                  0.07613    0.3345     0.8902        2.217   27.19
         smoothness_se compactness_se concavity_se concave.points_se
842302        0.006399        0.04904      0.05373           0.01587
842517        0.005225        0.01308      0.01860           0.01340
84300903      0.006150        0.04006      0.03832           0.02058
84348301      0.009110        0.07458      0.05661           0.01867
84358402      0.011490        0.02461      0.05688           0.01885
843786        0.007510        0.03345      0.03672           0.01137
         symmetry_se fractal_dimension_se radius_worst texture_worst
```

```
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
84358402     0.01756             0.005115        22.54         16.67
843786       0.02165             0.005082        15.47         23.75
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
84358402          152.20     1575.0           0.1374            0.2050
843786            103.40      741.6           0.1791            0.5249
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
84358402          0.4000               0.1625         0.2364
843786            0.5355               0.1741         0.3985
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
84358402                 0.07678
843786                   0.12440
```

```r
diagnosis <- wisc.df$diagnosis

nrow(wisc.data) #Q1
```

```
[1] 569
```

```r
table(diagnosis) #Q2
```

```
diagnosis
  B   M
357 212
```

```r
length(grep("mean", colnames(wisc.data), fixed = T)) #Q3
```

```
[1] 10
```

Q1. How many observatiosn are in this data set?

569

Q2. How many observations have a malignant diagnosis?

212

Q3. How many variables/features are suffixed with _mean?

10

Applying PCA to the Data

Before using PCA we first check to ensure the variances between each variable is consistent

```r
colMeans(wisc.data)
```

```
            radius_mean           texture_mean         perimeter_mean
           1.412729e+01           1.928965e+01           9.196903e+01
              area_mean        smoothness_mean       compactness_mean
           6.548891e+02           9.636028e-02           1.043410e-01
         concavity_mean    concave.points_mean          symmetry_mean
           8.879932e-02           4.891915e-02           1.811619e-01
 fractal_dimension_mean              radius_se             texture_se
           6.279761e-02           4.051721e-01           1.216853e+00
           perimeter_se                area_se          smoothness_se
           2.866059e+00           4.033708e+01           7.040979e-03
         compactness_se           concavity_se      concave.points_se
           2.547814e-02           3.189372e-02           1.179614e-02
            symmetry_se   fractal_dimension_se           radius_worst
           2.054230e-02           3.794904e-03           1.626919e+01
          texture_worst        perimeter_worst             area_worst
           2.567722e+01           1.072612e+02           8.805831e+02
       smoothness_worst      compactness_worst        concavity_worst
           1.323686e-01           2.542650e-01           2.721885e-01
   concave.points_worst         symmetry_worst fractal_dimension_worst
           1.146062e-01           2.900756e-01           8.394582e-02
```

```r
apply(wisc.data,2,sd)
```

```
            radius_mean            texture_mean          perimeter_mean
           3.524049e+00            4.301036e+00            2.429898e+01
              area_mean         smoothness_mean        compactness_mean
           3.519141e+02            1.406413e-02            5.281276e-02
         concavity_mean     concave.points_mean           symmetry_mean
           7.971981e-02            3.880284e-02            2.741428e-02
 fractal_dimension_mean               radius_se              texture_se
           7.060363e-03            2.773127e-01            5.516484e-01
           perimeter_se                 area_se           smoothness_se
           2.021855e+00            4.549101e+01            3.002518e-03
         compactness_se            concavity_se       concave.points_se
           1.790818e-02            3.018606e-02            6.170285e-03
            symmetry_se    fractal_dimension_se            radius_worst
           8.266372e-03            2.646071e-03            4.833242e+00
          texture_worst         perimeter_worst              area_worst
           6.146258e+00            3.360254e+01            5.693570e+02
       smoothness_worst       compactness_worst         concavity_worst
           2.283243e-02            1.573365e-01            2.086243e-01
   concave.points_worst          symmetry_worst fractal_dimension_worst
           6.573234e-02            6.186747e-02            1.806127e-02
```

Using PCA on the data set

```r
wisc.pr <- prcomp(wisc.data, scale = T)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
```

```
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

44.27% of the variance.

> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

3 principal components.

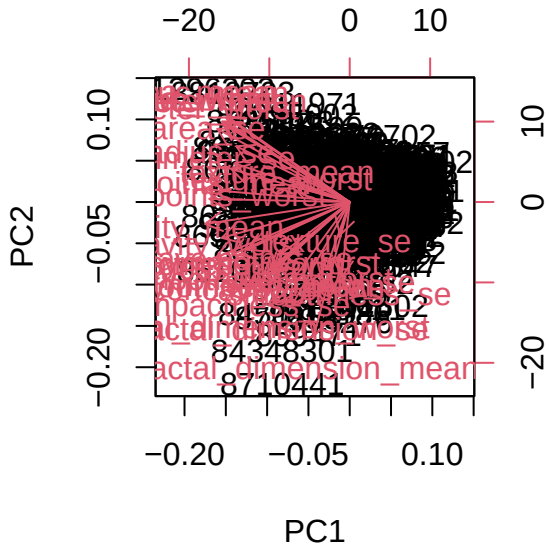
> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7 prinipal components

Visualizing PCA Results

First trying to visualize results using the `biplot()` function

```r
biplot(wisc.pr)
```



Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

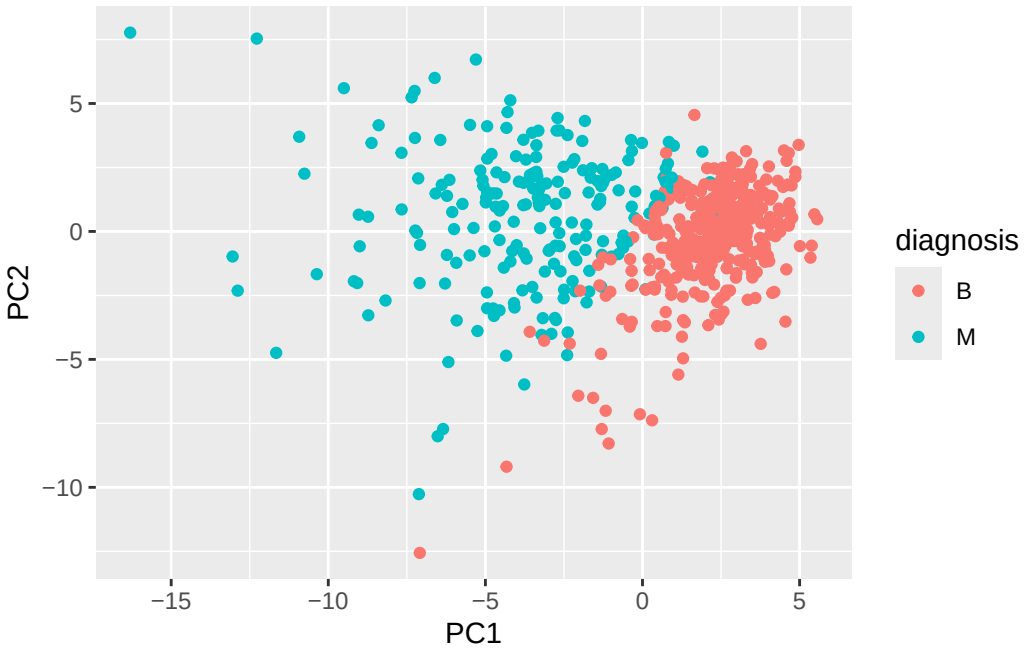
Since everything is scatterd messily and lablels are overlapping, there isnt anything that stands out

Using ggplot2 to plot PC1 and PC2

```
library(ggplot2)
```

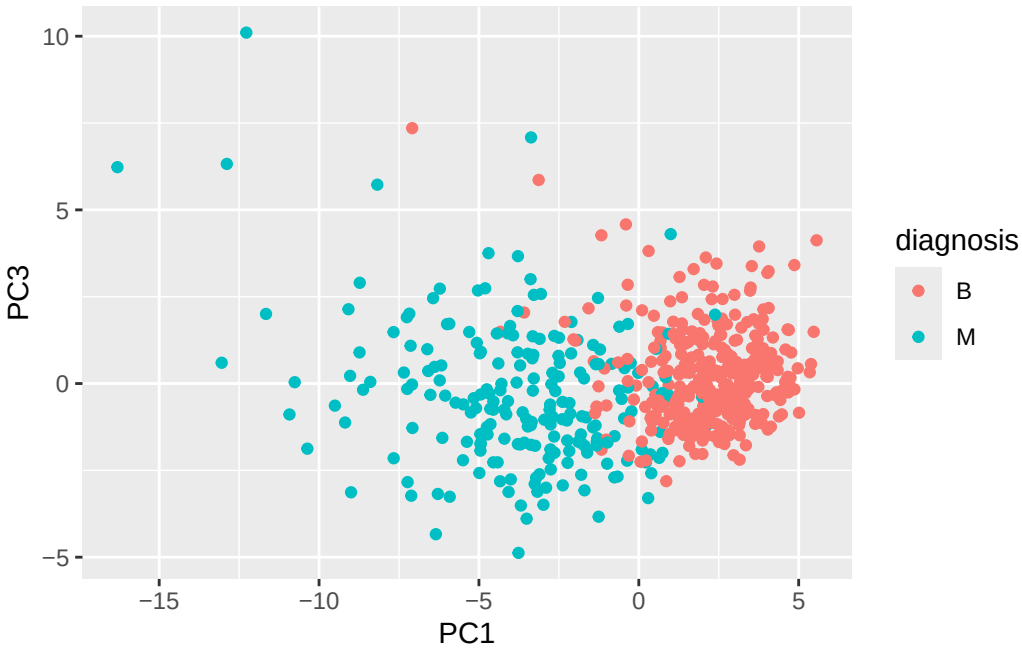
```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
pr_plot_1_2 <- ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(aes(col = diagnosis))
pr_plot_1_2
```



Further plotting PC1 and PC3

```
pr_plot_1_3 <- ggplot(wisc.pr$x) +
  aes(PC1, PC3) +
  geom_point(aes(col = diagnosis))
pr_plot_1_3
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

There are a clear clustering of points that represent the two doctor diagnoses.

Variance Explained

Identifying the scree point for principal component variance by plotting the variance by each successive component.

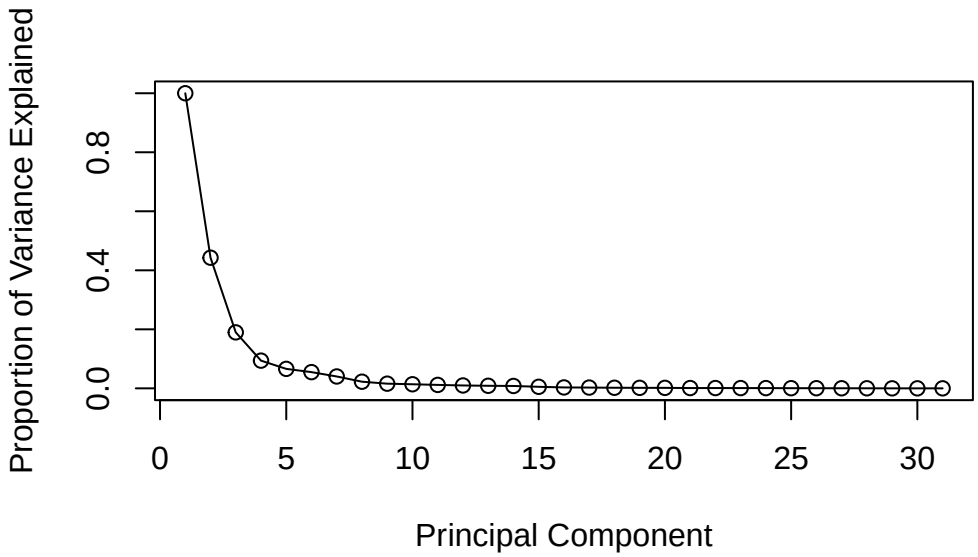
Creating a vector that contains the proportion of total variance in each PC.

```
pr.var <- wisc.pr$sdev^2
pve <- pr.var/sum(pr.var)
head(pve)
```

```
[1] 0.44272026 0.18971182 0.09393163 0.06602135 0.05495768 0.04024522
```

Plotting the variation across each added principal component

```
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



From the plot above, there is a clear scree point at the third principal component.

> Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

There are not, the other features have a lower contribution to the PC1 vector.

```
wisc.pr$rotation[,1]
```

```
            radius_mean            texture_mean          perimeter_mean
            -0.21890244             -0.10372458             -0.22753729
              area_mean         smoothness_mean        compactness_mean
            -0.22099499             -0.14258969             -0.23928535
         concavity_mean     concave.points_mean           symmetry_mean
            -0.25840048             -0.26085376             -0.13816696
 fractal_dimension_mean               radius_se              texture_se
            -0.06436335             -0.20597878             -0.01742803
           perimeter_se                 area_se           smoothness_se
            -0.21132592             -0.20286964             -0.01453145
         compactness_se            concavity_se       concave.points_se
            -0.17039345             -0.15358979             -0.18341740
```

```
           symmetry_se    fractal_dimension_se            radius_worst
           -0.04249842             -0.10256832             -0.22799663
         texture_worst         perimeter_worst              area_worst
           -0.10446933             -0.23663968             -0.22487053
      smoothness_worst       compactness_worst         concavity_worst
           -0.12795256             -0.21009588             -0.22876753
  concave.points_worst          symmetry_worst fractal_dimension_worst
           -0.25088597             -0.12290456             -0.13178394
```

```
max(abs(wisc.pr$rotation[,1]))
```

```
[1] 0.2608538
```

Heirarchical Clustering

First scale the data and assign distances between each point.

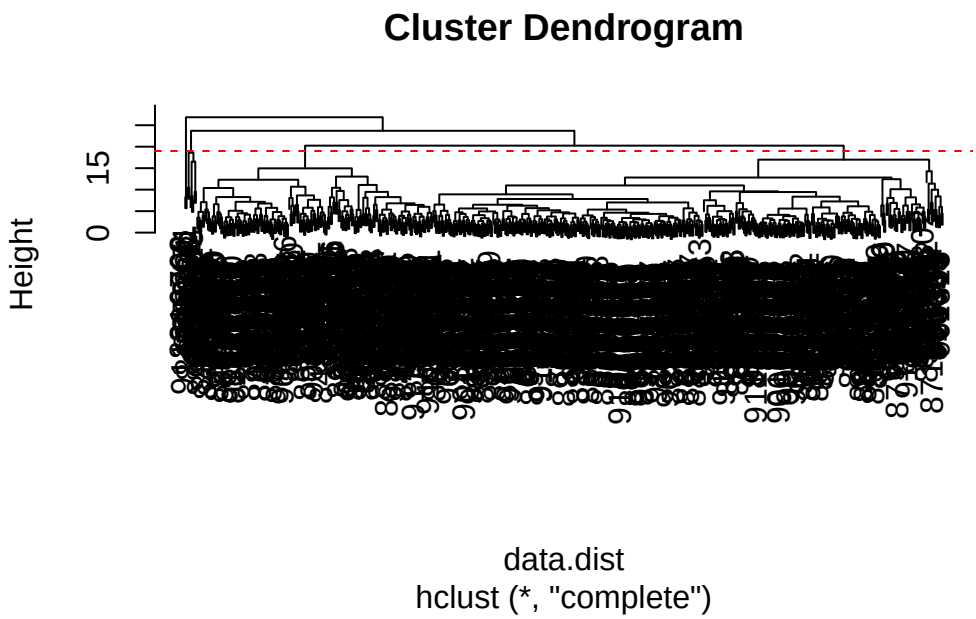
```
scaled.data <- scale(wisc.data)
data.dist <- dist(scaled.data)
```

Create a hierarchical clustering model using `hclust()`

```
wisc.hclust <- hclust(data.dist, method = "complete")
```

Creating a cluster deprogram using the `hclust()` result

```
plot(wisc.hclust)
abline(h = 19, col = "red", lty = 2)
```



Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

roughly h = 19

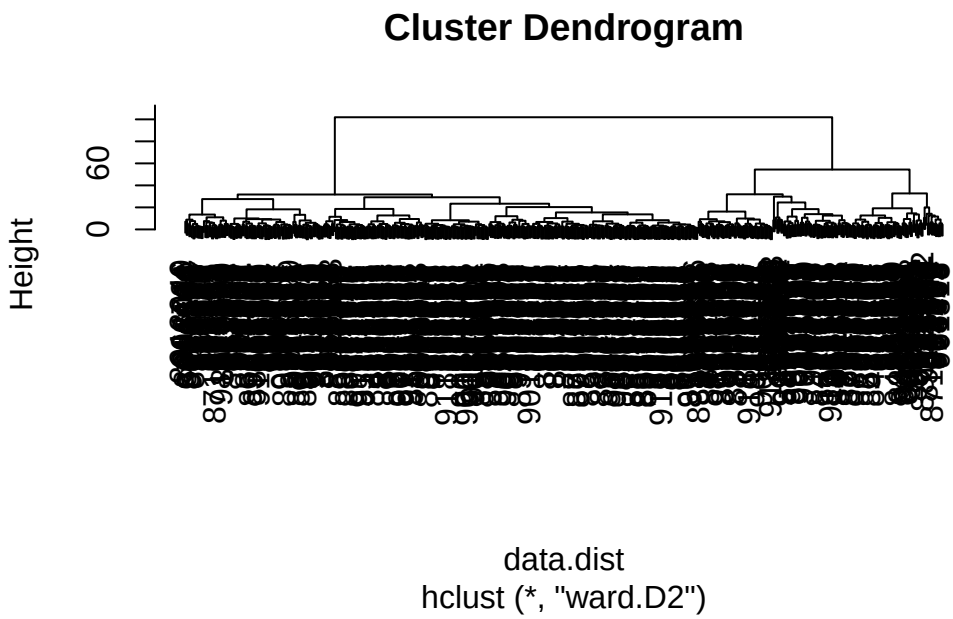
Using `cutree()` to create clusters of the data and compares to the clinically diagnosed clusters.

```
wisc.hclust.clusters <- cutree(wisc.hclust, h = 19)
table(wisc.hclust.clusters, diagnosis)
```

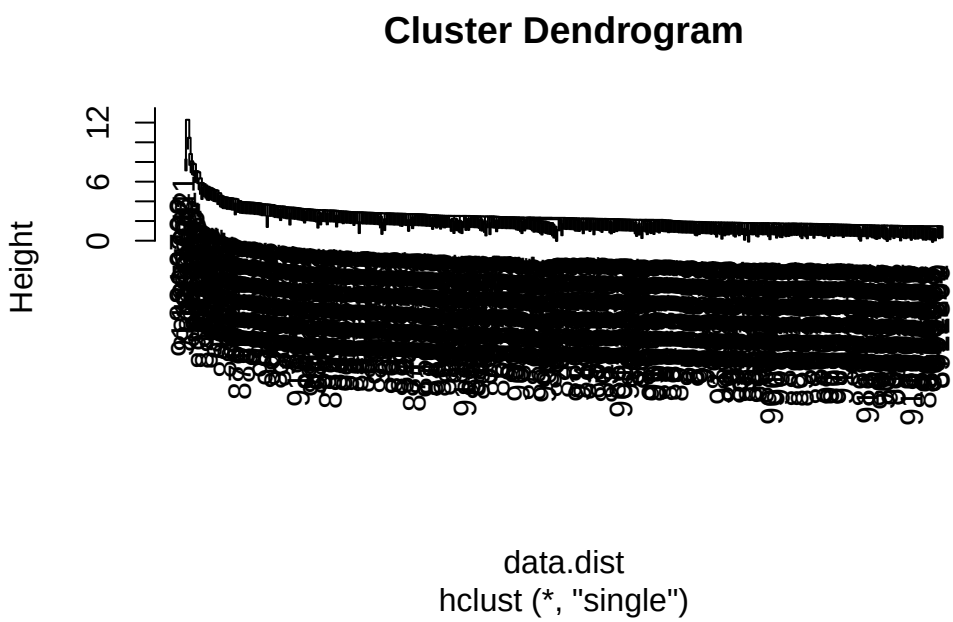
```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

Using different `method` arguments in `hclust()`

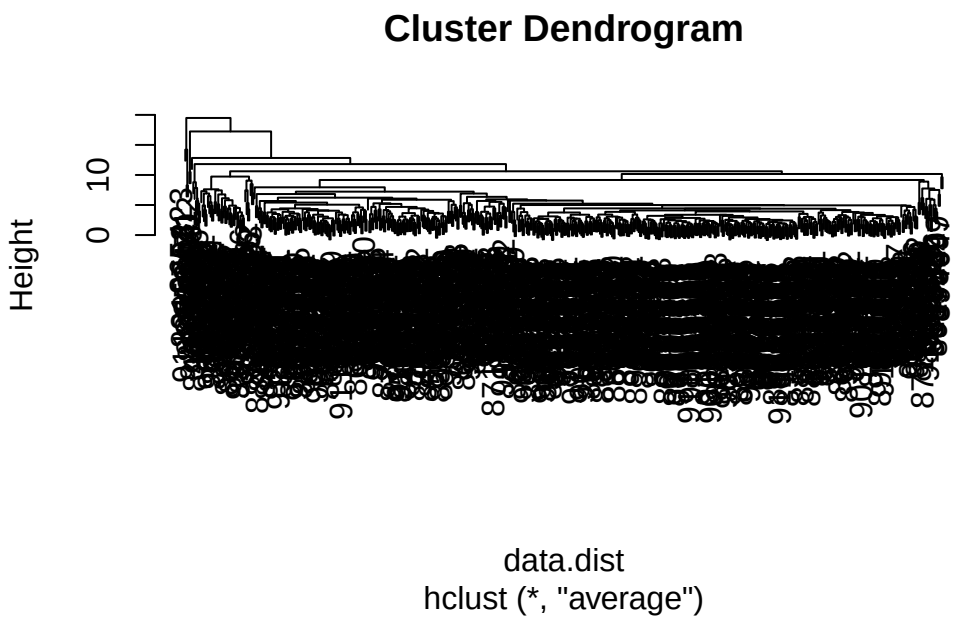
```
plot(hclust(data.dist, method = "ward.D2"))
```




```
plot(hclust(data.dist, method = "single"))
```



```r
plot(hclust(data.dist, method = "average"))
```



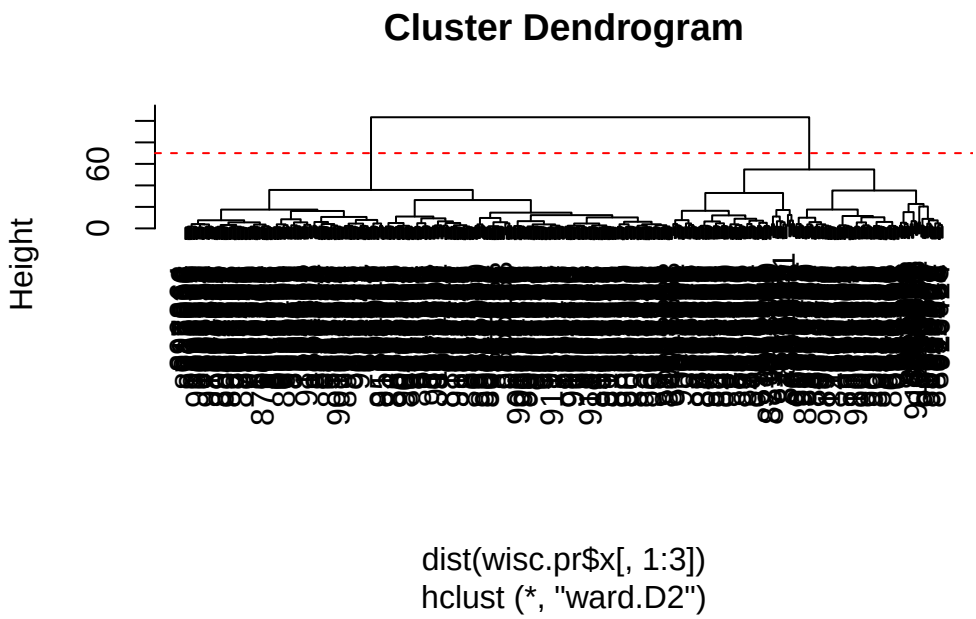
Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

The `ward.D2` method gives the cleanest results since it creates a plot with clear branches that show the clustering results. Furthermore with `ward.D2` it avoids small branches like in `complete`.

Combining Heirarchical Clustering with PCA

Using `hclust()` on the PCA data using enough components to account for 90% of the data. transformed the PCA data into distances from each point.

```r
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:3]), method = "ward.D2")
plot(wisc.pr.hclust)
abline(h = 70, col = "red", lty = 2)
```



Using `cutree()` to assign groups and comparing to the known diagnoses.

```
grps <- cutree(wisc.pr.hclust, k = 2)
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  24 179
   2 333  33
```

Cluster 1: 179 "M" diagnoses Cluster 2: 333 "B" diagnoses 179 true positives 24 false positives 333 true negatives 33 false negatives <- the worst case, try to minimize

Specificity Tp/(TP + FN)

```
179/(179+33)
```

```
[1] 0.8443396
```

Sensitivity TN/(TN + FP)

```r
333/(333+24)
```

```
[1] 0.9327731
```

> Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

Overall within the two groups that were assigned, the specificity of the groups was 0.84 and the sensitivity was 0.93 so `hclust()` model separates "M" and "B" well.

```r
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

Ignoring clusters 2 and 4: TP: 165 FP: 12 TN: 343 FN: 40

Specificity Tp/(TP + FN)

```r
165/(165 + 40)
```

```
[1] 0.804878
```

Sensitivity TN/(TN + FP)

```r
343/(343 + 12)
```

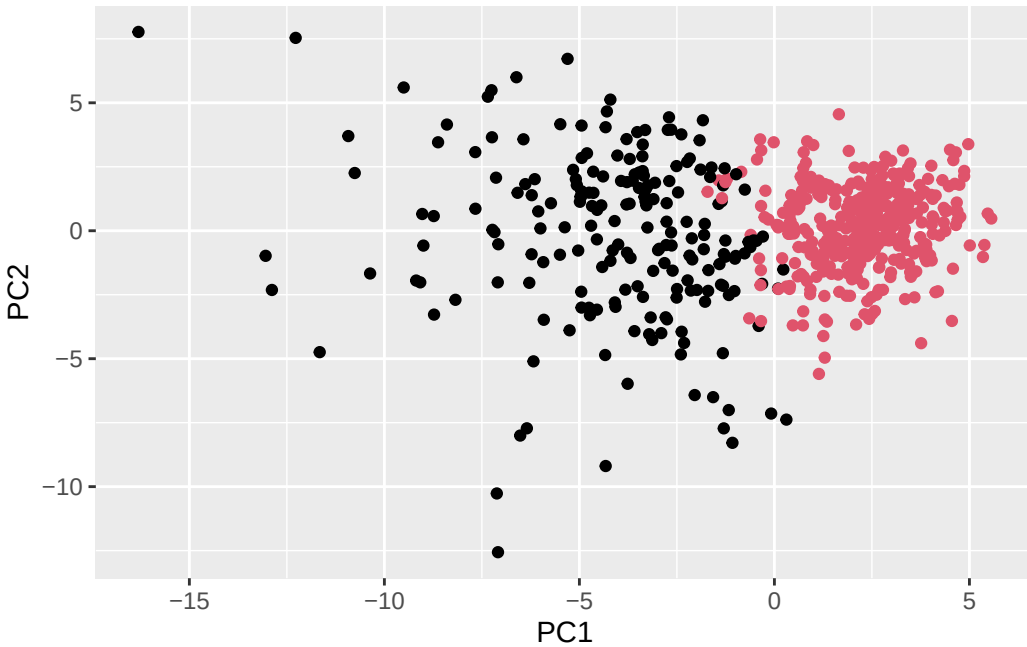
```
[1] 0.9661972
```

> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses?

Comparatively, the clustering off of the raw data results in a more messy result, creating 4 separate clusters with 2 clusters without clear distinctions. Numerically, the PCA model creates groups with higher specificity while the raw data creates models with higher sensitivity. Overall, with the messier groups using the raw data, the PCA groupings are more effective.

Adding groups to the PC1 vs PC2 plot.

```r
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col = grps)
```



Predicting Diagnoses using PCA results

Using the groups from the PCA results, we can predict the potential diagnoses by projecting them on the PCA space.

Importing new samples

```r
new <- read.csv("new_samples.csv")
```

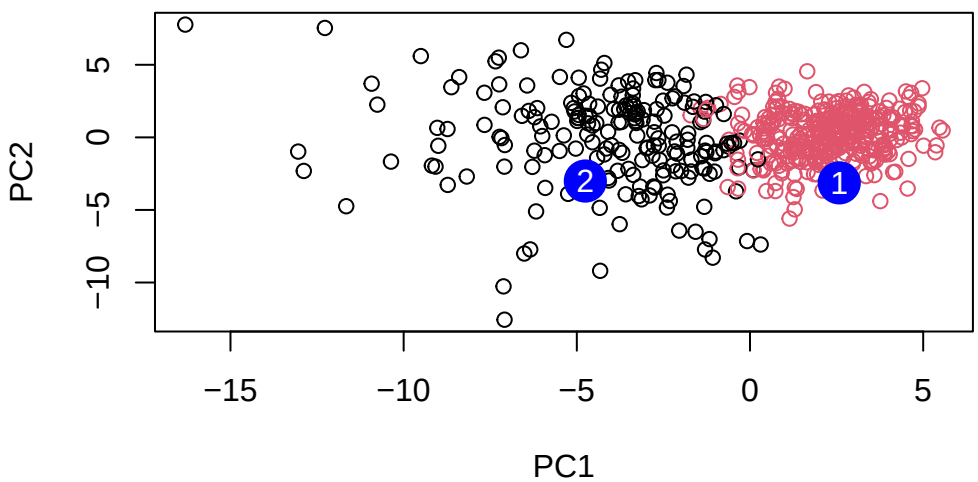
Using the `predict()` function, we can use the previous PCA ouput to assign the new samples PC scores using the PCs from the PCA.

```r
npc = predict(wisc.pr, newdata = new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
```

```
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

We should follow up on patient 1 because patient 1's tumor is within the group that has a high likelihood of being malignant.